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Roll No

EC-5004-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Communication Networks and Transmission Lines

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume suitable data, if required.

1. a) Discuss the iterative impedance and characteristic impedance of symmetrical two port network.
b) Explain the design of symmetrical attenuators. Also discuss their working principle.
2. a) Discuss briefly about different Matching Techniques.
b) Discuss about Symmetrical and Asymmetrical attenuators and their design.
3. a) Give an analysis and design of M-derived High-pass Filter.
b) Explain Butterworth Approximation Method for designing Low-pass Filter.
4. Discuss about butterworth approximation in filter design.
5. a) Discuss about foster and causer network.
b) Discuss about Bott-Duffin method.

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6. a) Realize the given function in Caur form-II

$$Z(s) = \frac{6 + 8s^2 + 2s^4}{s^3 + 2s}$$

- b) Derive the conditions required for a distortion less line.
7. a) Discuss about current and voltage on a dissipation less line. Also explain what is standing wave and standing wave ratio.
- b) What is the need of stub matching explain single and double stub matching.
8. Write short notes on (any two):
- a) Lattice and Bridged T networks
 - b) Brune's method
 - c) Design of two wire line and coaxial cable
 - d) Infinite line
